Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

AI/ML Engineer with 7+ years of software engineering experience, specializing in **production AI systems, RAG pipelines, and LLM orchestration**. At Intuit, owns the full 0→1 lifecycle -- partnering with stakeholders to prototype AI solutions, validating with golden-dataset eval harnesses, and hardening into enterprise-scale services on AWS. Built **AI on FHIR**, a clinical NLP API translating English/Spanish queries into FHIR data calls with HIPAA-style RBAC -- and **Leasel**, an 8-agent RAG system with hallucination mitigation and human escalation (93.6% precision across 1,240 test cases). Experienced with foundation model APIs (Claude, GPT, Llama), LangGraph agentic workflows, vector databases, and **HIPAA / healthcare data privacy** standards. Strong Python, FastAPI, SQL, AWS, Docker, Kubernetes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / FastAPI / SQL | LLM APIs (Claude, GPT, Llama) | RAG Pipelines / Vector Databases | AI Orchestration (LangGraph)

Prompt Engineering / Tool Calling | Hallucination Mitigation / AI Evals | LLM-as-Judge / Golden Dataset Testing

HIPAA / Healthcare Data Privacy | FHIR / Clinical NLP APIs | REST APIs / Backend Systems | AWS (Lambda, S3, RDS, SQS)

Docker / Kubernetes / Terraform | CI/CD (GitHub Actions) | Observability (Prometheus, OpenTelemetry)

Human-in-the-Loop Escalation | 0→1 Prototype to Production

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- AI/LLM Tools & Production Systems -- Virginia, USA

- **Forward-deployed AI delivery:** work directly with business stakeholders to probe operational bottlenecks, prototype solutions using Python/FastAPI + LLM APIs, and determine whether simple automation, rule-based logic, or generative AI is the right approach.
- Own the **0→1 lifecycle**: build working prototypes, validate with golden-dataset eval harnesses (LLM-as-judge, hallucination detection), instrument with Prometheus and OpenTelemetry, and harden into enterprise-scale services on AWS.
- Build **LLM-powered tools** using the Anthropic Messages API and Agent SDK -- prompt engineering, structured output, tool use, guardrails, and input validation for regulated financial data workflows.
- Implement **LangGraph agentic orchestration** with human-in-the-loop escalation; balance tradeoffs across model accuracy, latency, maintainability, and inference cost.
- Build **ML pipelines** (scikit-learn, XGBoost) from feature engineering through scoring APIs; deploy via Docker and GitHub Actions CI/CD.

Entrata -- Residential & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- Built and maintained **data-intensive backend systems** handling sensitive resident PHI -- Libsodium encryption for AWS S3 document storage, role-based access control, and multi-tenant data isolation aligned with privacy governance standards.
- Designed **Docker microservices architecture** (50% better scalability, 30% faster deployments); deployed on AWS with Terraform IaC; instrumented for production observability.
- Owned **REST API integrations** with external federal data sources; optimized 200+ SQL queries via execution plan analysis (30% performance gain).
- Built Python/Bash automation eliminating 80% of manual operational workload; reviewed 300+ PRs; mentored junior engineers.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built backend modules (Python, Java, PostgreSQL) with data import pipelines processing 1.6M records; implemented role-based access control and multi-tenant data isolation.

PROJECTS

AI on FHIR -- Clinical NLP & Healthcare API Integration

Python · Flask · React · TypeScript · FHIR · BERT · HIPAA · Docker

RESTful API that translates English/Spanish clinical queries into FHIR data calls using LLMs and BERT embeddings -- cutting manual clinical lookup time by ~60%. Fine-grained RBAC for HIPAA-style PHI data isolation, containerized deployment, and automated testing. Directly demonstrates healthcare informatics integration, data privacy governance, and clinical workflow automation.

github.com/venkatalakshmiparimala/AI_on_FHIR

Leasel -- LLM-Based Compliance Tool with Hallucination Mitigation

Python · LangGraph · Agent SDK · FastAPI · RAG · Eval Harness

8-agent LangGraph orchestration system: triage, RAG retrieval, rule evaluation, confidence scoring, and **human escalation at defined thresholds**. Hallucination mitigation built in: answers cited to exact source clauses; guardrails route low-confidence outputs to human review. LLM-as-judge CI harness: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- AI Evaluation & Multi-Provider Routing Infrastructure

Python · FastAPI · Provider Routing · Cost Management · CI/CD

15-endpoint FastAPI service with provider routing across five LLM/ML inference backends. Scoring engine evaluates output quality on five metrics (factual accuracy, hallucination rate, retrieval quality, completeness, compliance) with cost-aware routing; CI regression against golden datasets.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017